

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PCE 203 — Past Questions & Materials

by Renance

Course: PCE 203

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 200

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Paper 1 of 1 — FUPRE -PCE-203 - Introduction to the petrochemical Industry

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY * uples Pa

DEPARTMENT OF PETROCHEMICAL ENGINEERING

First Semester Examination 2024/2025 Session —

COURSE TITLE: Introduction to the Petrochemical Industry Credit Unit: 2

COURSE CODE: FUPRE-PCE-203 Time Allowed: 2 hrs

INSTRUCTIONS: Answer FOUR (4) questions only. Start each question on a fresh page. All questions carry equal marks.

1. (i) Give a concise definition of a petrochemical industry,

(11) State two (2) reasons why, most times, a petrochemical company is situated in the same place with a

crude oil refining company.

(111) List four (4) key petrochemical products that usually serve as feedstocks of other industrial processes, ad

(1) Briefly discuss the evolution of the petrochemical industry from “the early beginnings (1800s-1920s)” to “the rise of petrochemicals (1920s-1950s)”

(ii)State and explain the three (3) areas that will inform the future outlook of the petrochemical industry.

(iii Name three (3) exantples of majgr petrochemical companies’ industries in Nigeria and

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3. (i) State and briefly’explain the four (4) challenges which confronted the petrochemical industry in

outside Nigeria,

recent times, and equally paved the way for new opportunities. yee ot

(11) | \plain the following terms “Catalytic Cracking” and “Thermal Cracking”.

(41) State and explain four (4) additional skills and competence required of a petrochemical engineer for “ (

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addiessing emerging challenges and opportunities in the current labour market. “

sates

4. (i) State the three (3) types of crude oil. based on the hydrocarbon compound that are mostly prevalent in

them.

(ti) Define API Gravity and state the three (3) types of crude oil, based on their AP] gray ity.

(ii) State the nine (9) compounds prevalent in refinery crude. which are essentially targeted in the Various chemical and physical processes in the refinery .

1 of 2

| omic npounds:

5. (i) Name the following chemical comp

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(ii) Briefly explain "alkylation reaction" and State which of the above compounds is/are the target of the alkylation process.

(ii) Briefly explain “reforming reaction”

target of the reforming

and state which of the above

Compounds is/are the

process,

6. (1) Define "crude oil assay" and Slate two (2) usefulness of a crude oil 4

ssay?

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(1) State and briefly explain the five (5) basic types of petroleum refining

Operations.

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